

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – Industry Problem Solving Task

|                  |                                                                                                                                   |               |                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------|
| Course Code:     | DSA03                                                                                                                             | Course Title: | Natural Language Processing                       |
| CO Assessed:     | CO4 – Precision                                                                                                                   | Assessment:   | Assessment Tool 2 – Industry Problem Solving Task |
| Weightage:       | 40% of CIA                                                                                                                        |               |                                                   |
| CO5 Statement:   | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) |               |                                                   |
| Student Name:    | S Santhosh Reddy                                                                                                                  | Reg. No.:     | 192425017                                         |
| Section / Batch: | B.Tech AI &ML                                                                                                                     | Date:         | 08-08-2026                                        |

### Code of Conduct

I \_\_\_\_\_S Santhosh Reddy-192425017\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_S Santhosh Reddy\_\_\_\_

### Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

| Section / Topic                                                                                                | Focus                                                                                                                                                                                               | Question No. |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Representing Meaning & Meaning Structure of Language                                                           | Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications. | Q1           |
| Syntax-Driven Semantic Analysis                                                                                | Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.                                       | Q2           |
| First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis | Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.          | Q3           |

## Q1. AI-Powered Banking Customer Support Chatbot

A banking company is developing an AI-powered customer support chatbot to understand and respond to customer queries. The chatbot uses Context-Free Grammar (CFG) for syntactic parsing. However, it faces difficulties with ambiguous sentences, subject–verb agreement, and long conversational queries. The given query is “Show me the transactions with the card from last month.”

### a) Analysis of Ambiguity

The given sentence has both syntactic and semantic ambiguity. The phrase “with the card” can be interpreted in different ways. It may indicate transactions performed using a particular card, or it may refer to transactions associated with a card. The phrase “from last month” also needs to be correctly connected with the transactions.

A basic CFG can identify grammatical components such as noun phrases, verb phrases, and prepositional phrases. However, it cannot completely determine which interpretation is intended by the customer.

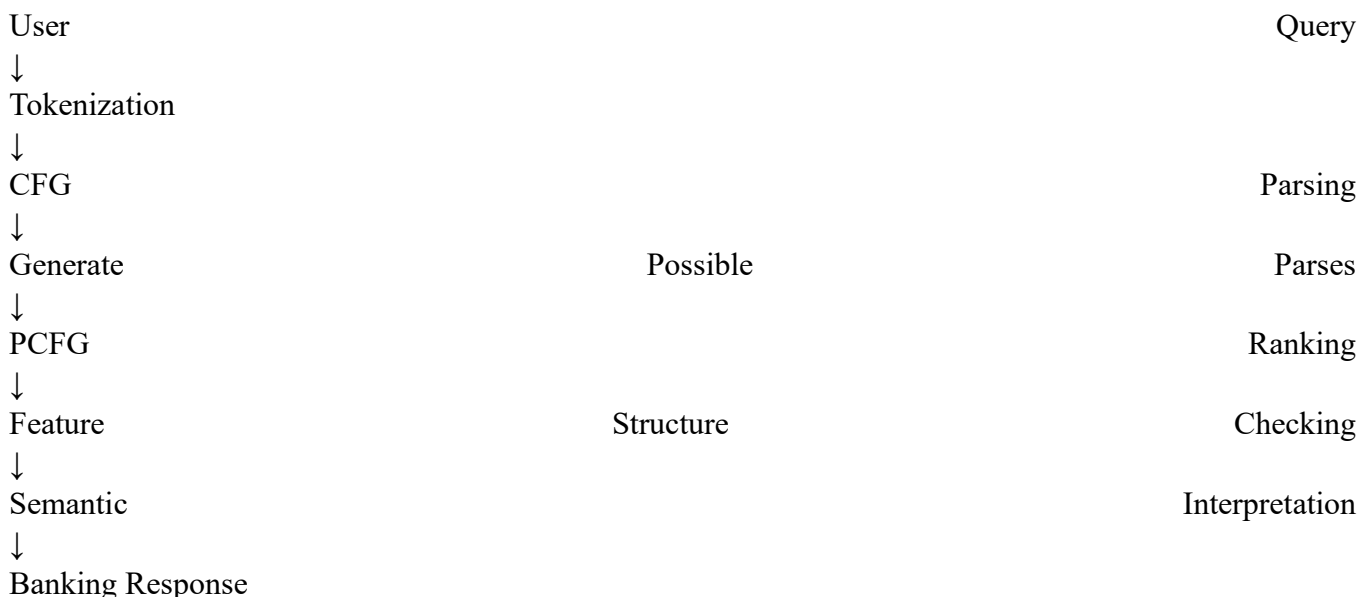
### b) Limitations of CFG

CFG is useful for representing the basic syntactic structure of sentences, but it has several limitations in real-world banking applications.

- It mainly focuses on syntax rather than complete semantic understanding.
- It may generate multiple parse trees for ambiguous sentences.
- It does not naturally represent contextual information.
- Handling subject–verb agreement requires additional mechanisms.

### c) Proposed Improved Solution

A hybrid parsing approach using CFG, PCFG, feature structures, and Earley parsing can be implemented. CFG can first construct the possible syntactic structures. PCFG can then assign probabilities to these structures and select the most likely interpretation.



### d) Industry Benefits

The proposed approach can improve the chatbot in several ways. It reduces incorrect interpretations and provides more reliable responses to customers.

## Q2. Voice Assistant System

A voice assistant system processes spoken commands using parsing techniques. The existing system uses top-down parsing but experiences backtracking problems, difficulty with incomplete or ambiguous inputs, and challenges in providing real-time responses. The example command is “Book a flight to Delhi with a window seat.”

### a) Analysis of Multiple Interpretations

The given command contains several important components. “Book a flight” represents the main action, “Delhi” represents the destination, and “window seat” represents the seating preference.

The semantic representation can be written as:

During parsing, the system must correctly identify the relationship between the different phrases. If the relationships are incorrectly identified, the voice assistant may produce an incorrect booking request.

### b) Limitations of Top-Down Parsing

Top-down parsing begins with the start symbol and attempts to generate a suitable parse for the input. When several grammar rules can be applied, the parser may need to backtrack and try alternative possibilities.

The major limitations are:

- Excessive backtracking for ambiguous commands.
- Increased processing time for complex sentences.
- Difficulty with incomplete spoken input.
- Poor handling of unexpected conversational expressions.
- Reduced suitability for real-time applications.
- Multiple possible interpretations may delay the final response.

These limitations are important because voice assistants are expected to respond quickly to spoken commands.

### c) Use of Earley Parsing

Earley parsing is suitable for handling ambiguous and complex sentences. It maintains partial parsing information while processing the input. Instead of repeatedly restarting the parsing process, it keeps track of possible structures.

### d) Comparison of Top-Down and Earley Parsing

Top-down parsing is relatively simple and works well with simple and predictable grammatical structures. However, its performance can decrease when the input is ambiguous or requires extensive backtracking.

Earley parsing is more flexible and can handle ambiguous, incomplete, and complex inputs. It maintains different possible states during parsing and can therefore be more suitable for conversational applications.

For a real-time voice assistant, Earley parsing provides better support for ambiguity and partial input handling. Hence, it is a suitable alternative to basic top-down parsing for this application.

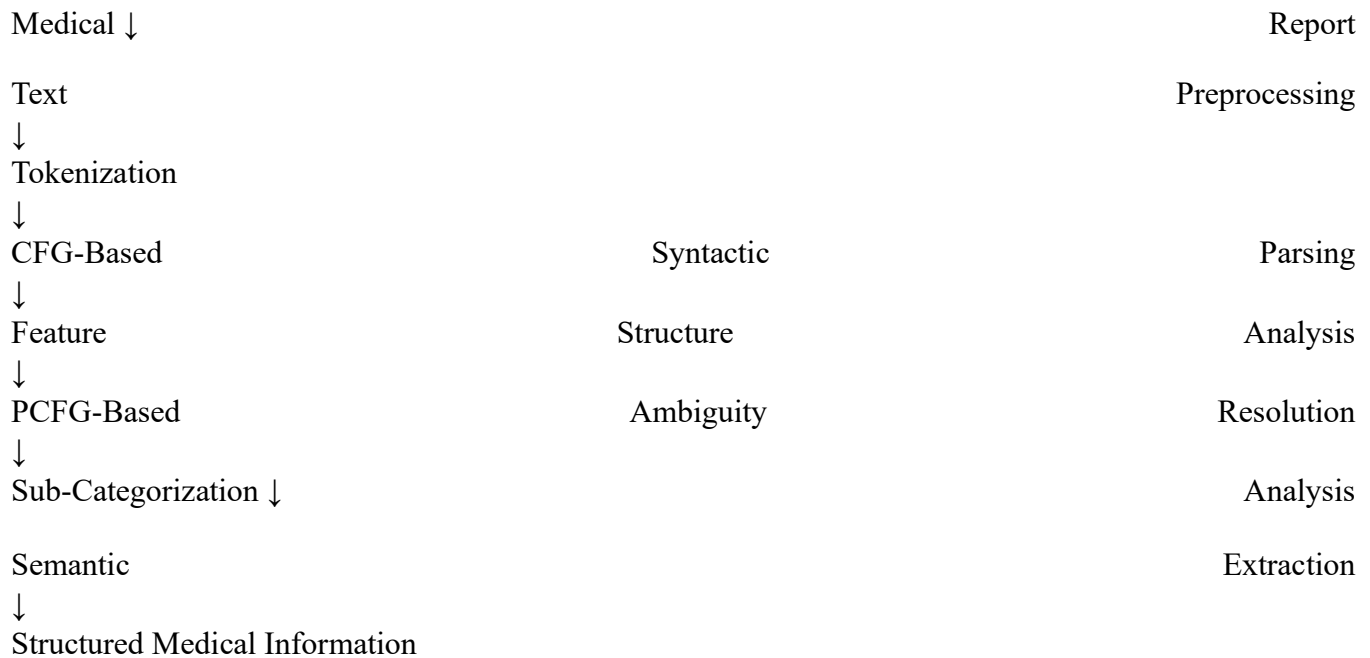
## Q3. Healthcare Medical Report Analysis

A healthcare company is developing an NLP system to analyze patient medical reports and automatically extract important information such as diagnosis, prescribed treatment, and follow-up actions. The system must handle complex medical sentences, structural ambiguity, agreement, relationships between medical

entities and actions, and real-time processing. The example sentence is “The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

### a) Complete NLP Architecture

A suitable architecture for the healthcare system is:



CFG provides the basic syntactic structure of the medical sentence. PCFG helps select the most probable interpretation when more than one parse is possible. Feature structures represent linguistic properties and help maintain agreement. Efficient parsing techniques support faster processing of complex medical reports.

### b) Step-by-Step Processing

Step 1: Input the medical report into the NLP system.

Step 2: Preprocess the text by separating words, phrases, and relevant linguistic units.

Step 3: Apply CFG to determine the syntactic structure of the sentence.

Step 4: Use feature structures to identify grammatical relationships and agreement.

Step 5: Apply PCFG to resolve possible structural ambiguities using probabilities.

Step 6: Identify important medical entities such as doctor, patient, medication, visit, and location.

Step 7: Apply sub-categorization frames to determine relationships between medical actions and their arguments.

Step 8: Convert the extracted information into structured output for further medical processing.

### c) Structured Output

For the given sentence, the system can produce the following information:

|           |            |            |     |            |
|-----------|------------|------------|-----|------------|
| Doctor:   |            | The        |     | doctor     |
| Previous  | action:    | Reviewed   | the | patient    |
| Time      | reference: | Last       |     | week       |
| Treatment | action:    | Starting   |     | medication |
| Follow-up | action:    | Scheduling | a   | follow-up  |
|           |            |            |     | visit      |

Location: Chennai

The sentence does not explicitly provide a diagnosis, so the system should not generate a diagnosis that is not present in the input.

#### **d) Sub-Categorization Frames**

Sub-categorization helps identify the arguments required by medical verbs. For example:  
recommend(Doctor, Action)

Here, the doctor is the entity performing the action, while the recommended action is the starting of medication and scheduling of a follow-up visit.

This approach helps the system understand relationships instead of simply identifying individual words. It is particularly useful for medical reports where the relationship between a patient, doctor, treatment, and follow-up action is important.

#### **e) Real-Time Processing**

For hospital applications, the system should process medical reports quickly. Efficient parsing techniques can reduce unnecessary computation. PCFG can be applied for resolving important ambiguities, while feature structures can provide grammatical consistency. Processing can also be divided into smaller stages so that information can be extracted without waiting for the entire hospital database to be processed.

#### **f) Accuracy and Scalability**

Accuracy can be improved by combining syntactic, probabilistic, and semantic techniques rather than depending only on CFG. Medical-specific vocabulary and sub-categorization rules can help identify relationships between medical actions and entities.

For scalability, the system should use efficient parsing methods and modular processing. This allows the same architecture to process increasing numbers of medical reports while maintaining reasonable response time. The proposed combination of CFG, PCFG, feature structures, and efficient parsing addresses the major requirements of the healthcare application.

#### **Conclusion**

CFG provides the basic syntactic foundation, while PCFG helps resolve ambiguity, feature structures handle linguistic relationships, and efficient parsing improves processing speed. Combining these techniques provides a stronger semantic processing framework for banking chatbots, voice assistants, and healthcare NLP systems.

## Rubrics for Calculation

| Criteria                            | Weight age | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                                |
|-------------------------------------|------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|----------------------------------------------------|
| Understanding of Industry Problem   | 20%        | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding ; misinterprets problem context |
| Application of Engineering Concepts | 25%        | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts            |
| Solution Design                     | 25%        | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                    |
| Use of Tools                        | 15%        | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                  |
| Analysis                            | 10%        | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                                |
| Professionalism                     | 5%         | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation                |

| Q. No. / Criteria Level | Understanding of Industry Problem | Application of Engineering Concepts | Solution Design | Use of Tools | Analysis | Professionalism | Sub. Total | Total % |
|-------------------------|-----------------------------------|-------------------------------------|-----------------|--------------|----------|-----------------|------------|---------|
|-------------------------|-----------------------------------|-------------------------------------|-----------------|--------------|----------|-----------------|------------|---------|

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| Faculty In-charge               | Course Coordinator              | Program Director                |
|---------------------------------|---------------------------------|---------------------------------|
| Signature: _____<br>Date: _____ | Signature: _____<br>Date: _____ | Signature: _____<br>Date: _____ |